SteelVision:

A CNN-Based Deep Learning Framework for Automated Steel Surface Defect Detection with Streamlit Deployment

Abstract

Steel surface defect detection is a critical task in modern manufacturing industries where product quality, safety, and cost efficiency are directly affected by inspection accuracy. Traditional manual inspection methods are slow, subjective, and prone to human error. This paper presents **SteelVision**, a Convolutional Neural Network (CNN) based automated defect detection system designed to identify surface defects on steel materials with high accuracy and real-time performance. The proposed framework leverages a structured dataset containing annotated training and validation images to learn discriminative defect patterns. The trained model is deployed using Streamlit to provide an interactive and user-friendly inspection interface suitable for industrial environments. Experimental results demonstrate strong classification performance, efficient inference speed, and practical deployment feasibility, making SteelVision a reliable solution for intelligent steel quality inspection.

1. Introduction

Quality assurance in steel manufacturing is essential to ensure structural integrity, reduce production losses, and maintain industrial standards. Surface defects such as scratches, cracks, pits, and inclusions can significantly affect the mechanical properties and aesthetic quality of steel products.

Traditional inspection approaches rely heavily on human operators, which introduces several limitations:

- Inconsistent inspection accuracy
- Operator fatigue
- Low scalability
- High labor costs

With the advancement of deep learning, computer vision systems have shown remarkable success in automated defect detection. Convolutional Neural Networks (CNNs), in particular,

are highly effective at extracting spatial features from images and identifying complex surface patterns.

This research introduces **SteelVision**, a CNN-driven inspection framework that:

- Automatically detects steel surface defects
- Learns from annotated industrial datasets
- Provides real-time predictions
- Offers an interactive Streamlit deployment for industrial usability

2. Related Work

Recent research in automated defect detection has focused on deep learning approaches due to their superior feature learning capability compared to traditional image processing techniques.

Earlier machine vision methods relied on handcrafted features such as texture descriptors and edge detection, which struggled with illumination variations and complex defect patterns. Deep learning models, especially CNN architectures, have overcome these limitations by learning hierarchical feature representations directly from data.

Industrial defect detection research has explored:

- CNN classification models for defect presence detection
- Object detection models for localization
- Segmentation models for pixel-level defect analysis

However, many existing solutions lack real-time usability and practical deployment interfaces. SteelVision addresses this gap by combining accurate CNN modeling with an accessible Streamlit interface.

3. Dataset Description

The SteelVision system is trained using a structured dataset consisting of:

- **Training set:** 50,000 images
- **Validation set:** 5,000 images

- **Annotation files:** train.txt and val.txt containing image paths and labels

The dataset includes multiple categories of steel surface defects captured under industrial imaging conditions. The annotation files provide supervised learning guidance by mapping each image to its corresponding defect class.

3.1 Data Characteristics

- High-resolution industrial imagery
- Realistic illumination variations
- Multiple defect types
- Class distribution suitable for supervised CNN training

3.2 Data Preprocessing

To improve model robustness, the following preprocessing steps were applied:

- Image resizing
- Normalization
- Data augmentation (rotation, flipping, brightness variation)
- Train-validation split verification

4. Methodology

4.1 System Overview

SteelVision follows a deep learning pipeline consisting of:

1. Data preprocessing and augmentation
2. CNN model training
3. Model validation and optimization
4. Streamlit deployment for real-time inference

4.2 CNN Architecture

The proposed CNN model is designed to capture both low-level texture details and high-level defect patterns. The architecture includes:

- Convolutional layers for feature extraction
- Batch normalization for training stability
- ReLU activation for non-linearity
- Max-pooling layers for spatial reduction
- Fully connected layers for classification
- Softmax output for multi-class prediction

This architecture balances performance and computational efficiency, making it suitable for real-time industrial deployment.

4.3 Training Strategy

The model training process includes:

- Cross-entropy loss optimization
- Adaptive learning rate scheduling
- Early stopping to prevent overfitting
- Validation monitoring for performance evaluation

5. Streamlit Deployment

To ensure industrial usability, the trained SteelVision model is deployed using Streamlit, enabling:

- Image upload for inspection
- Instant defect prediction
- Confidence score visualization
- User-friendly industrial dashboard

This deployment transforms the research model into a practical inspection tool accessible without deep technical knowledge.

6.Experimental Results

The SteelVision system demonstrates strong performance across validation metrics.

6.1 Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score

6.2 Performance Analysis

The CNN model achieved:

- High classification accuracy on validation data
- Strong generalization across defect categories
- Fast inference suitable for real-time inspection

The results indicate that SteelVision effectively captures defect patterns while maintaining deployment efficiency.

7. Discussion

The proposed SteelVision framework offers several advantages:

- Reduced inspection time
- Consistent detection accuracy
- Scalable industrial deployment
- Minimal human intervention

However, performance may be influenced by:

- Extreme lighting variations

- Rare defect categories
- Dataset imbalance

Future improvements can include object detection localization, segmentation integration, and edge deployment optimization.

8. Conclusion

This paper presented **SteelVision**, a CNN-based automated steel surface defect detection framework with Streamlit deployment. The system successfully combines deep learning accuracy with real-world usability, providing a scalable and efficient alternative to manual inspection.

Experimental evaluation confirms that SteelVision delivers reliable defect classification performance while enabling interactive industrial deployment. The framework represents a significant step toward intelligent manufacturing and automated quality assurance.

9. Future Work

Future research directions include:

- Real-time video inspection integration
- Transformer-based defect detection models
- Edge device optimization for factory deployment
- Defect localization and segmentation enhancement